Factorization of Polynomials Over the Boolean Semifield

MAA Student Talk

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Background

The Boolean semifield, denoted as $\mathbb{B}$, is an algebraic structure with two operations: logical OR, analogous to addition, and logical AND, analogous to multiplication. $\mathbb{B}$ the only finite semifield that is not a field. Our research explores the factorization of polynomials over the Boolean semifield, denoted as $\mathbb{B}[x]$.

A key challenge in $\mathbb{B}[x]$ is that factorization is not unique. Specifically, we investigate the factorization of a particular class of polynomials in $\mathbb{B}[x]$ namely:

$$f_n(x) = \sum_{i=0}^{n} x^i = x^n + x^{n-1} + \cdots + x + 1$$

Understanding the non-unique factorization of these polynomials is a central focus of our study. This research contributes to a deeper understanding of algebraic structures beyond traditional fields and rings.

Definitions

- A semifield is an algebraic structure like a field with addition and multiplication. That does not necessarily have additive inverses and does have multiplicative inverses.
- Let $\mathbb{B} = \{0, 1\}$ be the Boolean semifield equipped with logical OR as addition and Logical AND as Multiplication.
- Let $\mathbb{B}[x]$ be the set of all polynomials in x over $\mathbb{B}$.
- Let $f_n(x) \in \mathbb{B}[x]$, be defined as $f_n(x) = x^n + x^{n-1} + \cdots + x + 1$.

Necessary but Not Sufficient Condition for Factorization

Theorem

Let $f(x) = \sum_{i=0}^{n} f_i x^i$ *with* $f_0 = 1$. *If* $f(x)$ *is factorable, then there exist* j, k *with* $1 \leq j, k \leq n-1$ *and* $j + k = n$ *such that* $f_j = f_k = 1$.

Proof.

Assume $f(x) = g(x)h(x)$ with $1 \leq \deg(g), \deg(h) \leq n-1$. Let $\deg(g) = j$, $\deg(h) = k$. Then $g(x) = x^j + g'(x) + 1, \quad h(x) = x^k + h'(x) + 1$.

$$\begin{aligned} f(x) &= g(x)h(x) \\ &= (x^j + g'(x) + 1)(x^k + h'(x) + 1) \\ &= x^{j+k} + x^j + x^k + \big(x^j h'(x) + x^k g'(x) + g'(x)h'(x) + g'(x) + h'(x)\big) + 1. \end{aligned}$$

Thus $j + k = n$ and $f_j = f_k = 1$ □

Remarks.

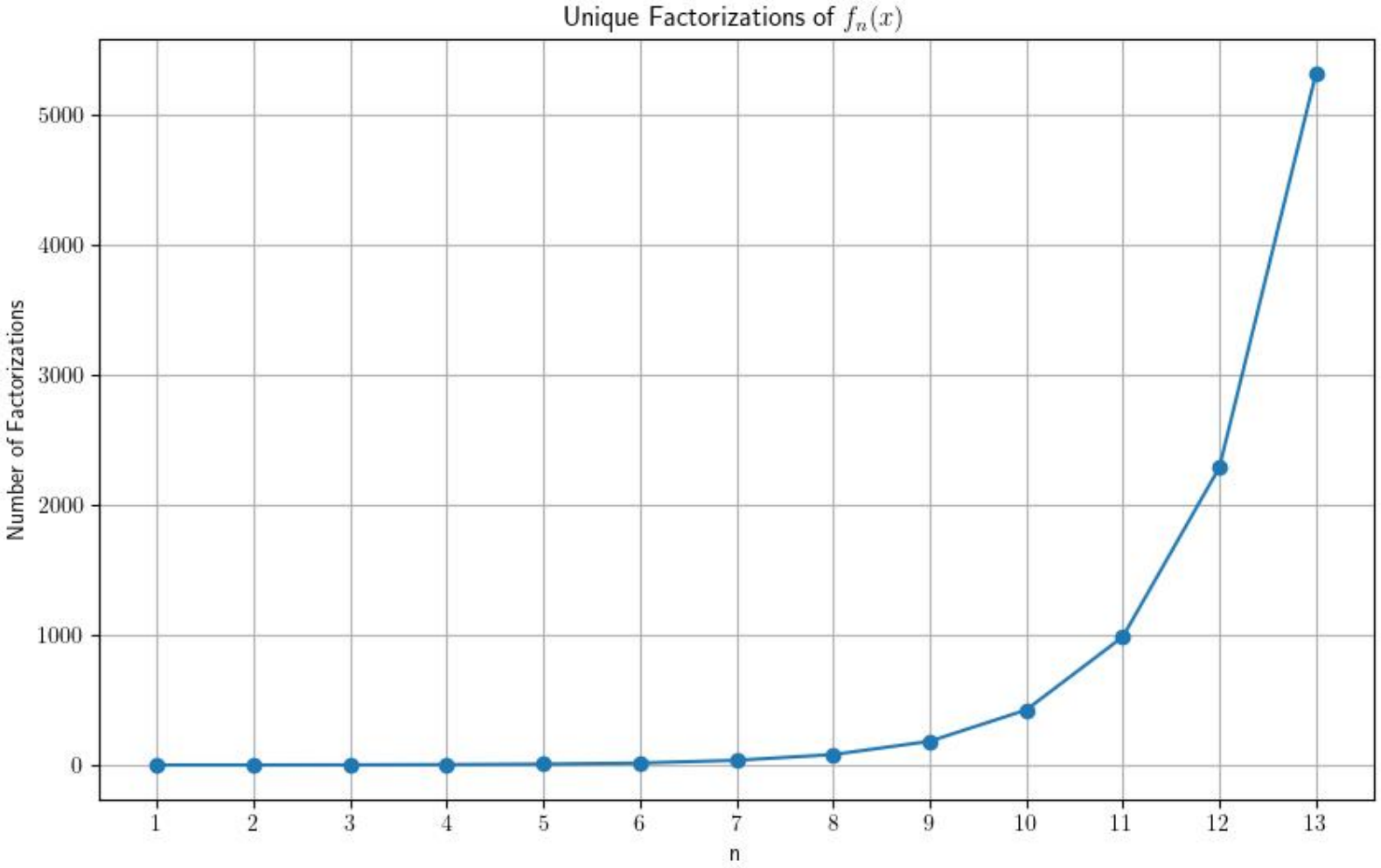
$f(x) = x^6 + x^4 + x^2 + x + 1$ satisfies the condition but is not factorable.

$f(x) = x^6 + x^5 + x^2 + 1$ does not satisfy the condition and is not factorable.

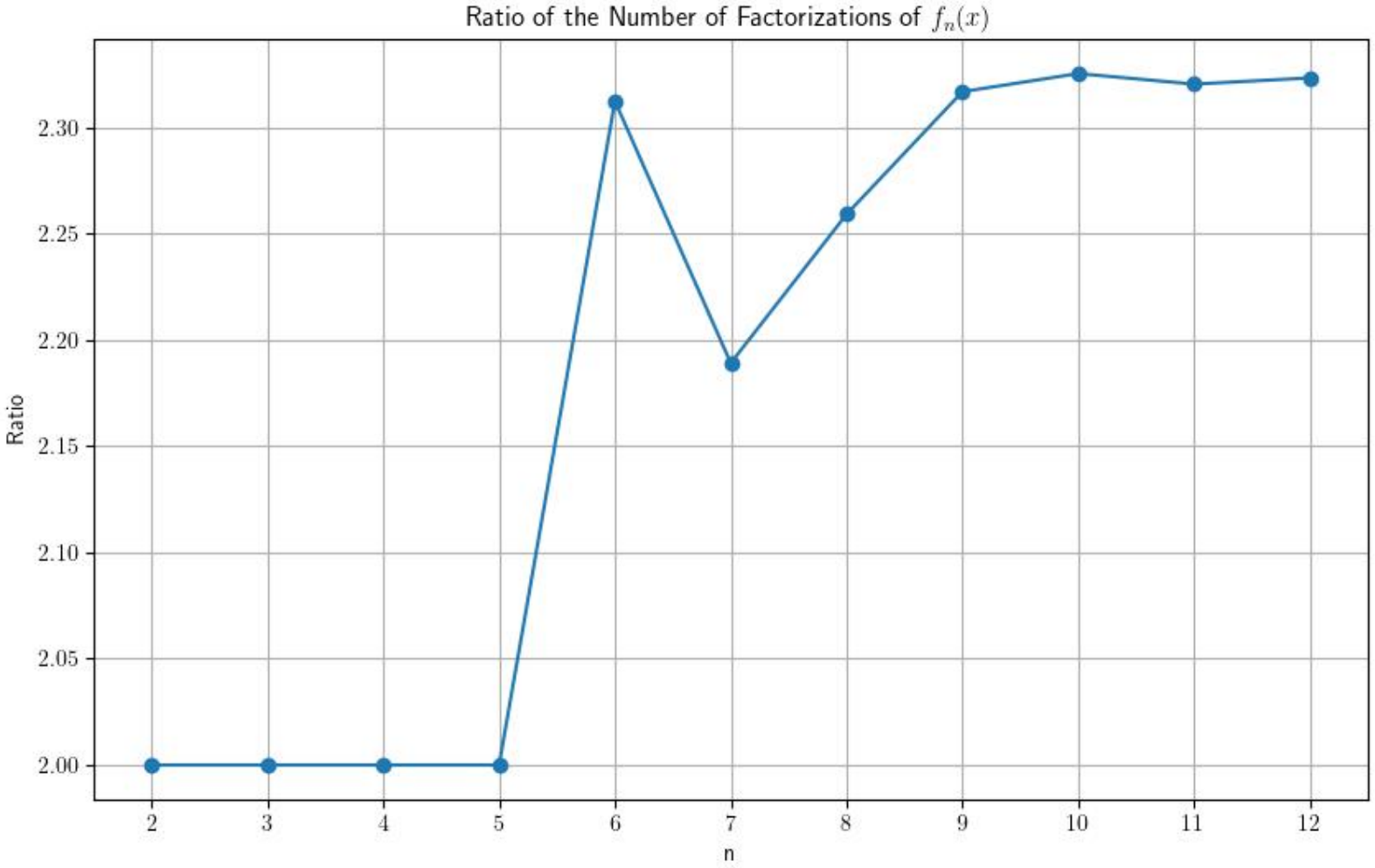
Examples of Possible Factorization

n	$f_n(x)$	Factorizations
2	$x^2 + x + 1$	$(x+1)^2$
3	$x^3 + x^2 + x + 1$	$(x+1)^3$ $(x+1)(x^2+1)$
4	$x^4 + x^3 + x^2 + x + 1$	$(x+1)^4$ $(x+1)^2(x^2+1)$ $(x+1)(x^3+x^2+1)$ $(x+1)(x^3+x+1)$
5	$x^5 + x^4 + x^3 + x^2 + x + 1$	$(x+1)^5$ $(x+1)^3(x^2+1)$ $(x+1)^2(x^3+x^2+1)$ $(x+1)^2(x^3+x+1)$ $(x+1)^2(x^3+1)$ $(x+1)(x^4+x^3+x^2+1)$ $(x+1)(x^4+x^2+x+1)$ $(x+1)(x^2+1)^2$

Exponential Growth of Factors



Growth Rate of Factors



Divisibility of $f_n(x)$

Lemma

Lemma 3.1 from the paper 'On the Sequence A079500 and Its Combinatorial Interpretations' [3] states conditions for divisibility in the numbral arithmetic: For each $n, d > 0$, it holds that $[d]$ is a divisor of $[2^n - 1]$ if and only if it ends with the digit 1, and it does not contain any subsequence of 0s having length greater than $n - k$, with k being the length of $[d]$.

Theorem

Let $g(x) \in \mathbb{B}[x]$ such that the constant term of $g(x)$ is 1. If $\deg(g(x)) = k$ and $2k - 1 \leq n$, then $g(x)$ divides $f_n(x)$.

Proof.

Let $\theta(h(x))$ be the isomorphism between $\mathbb{B}[x]$ and the numbral arithmetic $\theta(h(x)) = [h(2)]$. Assume $2k - 1 \leq n$ and let $d = \theta(g(x))$ (notice $[d]$ is length $k + 1$). Let m be the maximum length of zeros in $[d]$. Notice $m \leq k - 1$. Notice $\theta(f_n(x)) = 2^{n+1} - 1$. By the above lemma, d divides $2^{n+1} - 1$ because $k - 1 \leq n + 1 - (k + 1)$, which is equivalent to $2k - 1 \leq n$. And by applying θ^{-1} we see $g(x)$ divides $f_n(x)$. $\square$

Factorizations Of The Form $(x+1)h(x) = f_n(x)$

Theorem

$(x+1)h(x) = f_n(x)$ *if and only if* $h(x) = \sum_{i=0}^{n-1} h_i x^i$ *With* $h_{n-1} = h_0 = 1$ *and* $h_i + h_{i-1} = 1$ *for* $0 < i < n-1$.

Proof.

($\rightarrow$): Assume that $(x+1)h(x) = f_n(x)$ and let $h(x) = \sum_{i=0}^{n-1} h_i x^i$. (Notice $h_{n-1} = h_0 = 1$ otherwise it would contradict the assumption). Now assume for the sake of contradiction that there exists $k < n-1$ such that $h_k = h_{k-1} = 0$. Consider the following:

$$(x+1)h(x) = x\sum_{i=0}^{n-1} h_i x^i + \sum_{i=0}^{n-1} h_i x^i.$$

Notice the k term will have coefficient $h_k + h_{k-1} = 0$, contradicting the original assumption.
($\leftarrow$): Assume $h(x) = \sum_{i=0}^{n-1} a_i x^i$ with $h_{n-1} = h_0 = 1$ and for all h_i, $h_i + h_{i-1} = 1$. Consider the following:

$$(x+1)h(x) = \sum_{i=0}^{n-1} a_i x^{i+1} + \sum_{i=0}^{n-1} a_i x^i = a_{n-1}x^n + \sum_{i=1}^{n-1}(a_{i-1} + a_i)x^i + a_0 = f_n(x).$$

Therefore, the theorem holds. $\square$

Factorizations Of The Form $(x+1)h(x) = f_n(x)(Cont.)$

Theorem

Let $G_{n\,1} = \{h(x) | (x+1)h(x) = f_n(x)\}$ *Let* F_n *be the nth Fibonacci number.* $|G_{n\,1}| = F_n$ *for* $n \geq 1$

Proof (by Strong induction).

$G_{1\,1} = \{1\}, \quad |G_{1\,1}| = 1 = F_1 \quad \& \quad G_{2\,1} = \{x+1\}, \quad |G_{2\,1}| = 1 = F_2$
Therefore the base cases hold. Assume that $|G_{j\,1}| = F_j$ for all $j \in \{1, 2, \ldots, k\}$ for some k.
Let $h(x) \in G_{(k+1)\,1}$. Notice $\deg(h(x)) = k$ and $h_k = 1$ consider the possible values for h_{k-1}.

1. Case 1. $h_{k-1} = 1$. In which case $h(x) = x^k + h'(x)$ Where $h'(x) \in G_{k\,1}$.
2. Case 2. $h_{k-1} = 0$, So $h_{k-2} = 1$. In which case $h(x) = x^k + 0x^{k-1} + h''(x)$ Where $h''(x) \in G_{(k-1)\,1}$.

So, $|G_{(k+1)\,1}| = |G_{k\,1}| + |G_{(k-1)\,1}| = F_k + F_{k-1} = F_{k+1}$. Therefore, by the Principle of Mathematical Induction $|G_{n\,1}| = F_n$ for all $n \geq 1$. □

Factorizations Of The Form $(x^k+1)g(x) = f_n(x)$

Theorem

Let $G_{n\,k} = \{g(x) | (x^k+1)g(x) = f_n(x)\}$, *let* $m = n - 3k + 1$, *let* $q = \lfloor \frac{m}{k} \rfloor$, *and let* $r = m \bmod k$.

$$|G_{n\,k}| = (F_{q+3})^r \times (F_{q+2})^{k-r}$$

Sketch.

Consider $g(x) = \sum_{i=0}^{n-k} g_i x^i \in G_{nk}$.

$$\begin{aligned}(x^k+1)g(x) &= \sum_{i=0}^{n-k} g_i x^{i+k} + \sum_{i=0}^{n-k} g_i x^i \\ &= \sum_{i=k}^{n} g_{i-k} x^i + \sum_{i=0}^{n-k} g_i x^i \\ &= \sum_{i=n-k+1}^{n} g_i x^i + \sum_{i=k}^{n-k} (g_k + g_{i-k}) x^i + \sum_{i=0}^{k-1} g_i x^i = f_n(x)\end{aligned}$$

Notice this implies $g_i = 1$ for $i \leq k-1$ or $i \geq n-2k+1$ and $g_i + g_{i-k} = 1$ for $k \leq i \leq n-2k$

Factorizations Of The Form $(x^k + 1)g(x) = f_n(x)$ (Cont.)

Sketch.

Construct a partition the coefficients h_i where $k \leq i \leq n - 2k$. Let $g_i \in P_j$ if $i \equiv j \mod k$.

P_0	$g_k,\ g_{2k},\ \cdots$	$i \equiv 0 \mod k$
P_1	$g_{k+1},\ g_{2k+1}, \cdots$	$i \equiv 1 \mod k$
$\vdots$	$\vdots$	$\vdots$
P_0	$g_{2k-1},\ g_{3k-1}, \cdots$	$i \equiv k-1 \mod k$

Let $m = n - 3k + 1$, $q = \lfloor \frac{m}{k} \rfloor$, and $r = m \mod k$. The length of r of these partitions is $q + 1$ and $k - r$ have length q. It can be shown that the number of options (where $g_i + g_{i-k} = 1$) for P_j of length q is F_{q+2}. So, The number of valid polynomials is $(F_{q+3})^r \times (F_{q+2})^{k-r}$ $\square$

Sources

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Related Topics

1. Set Addition
   $A + B = \{a + b | a \in A \text{ and } b \in B\}$ where $A, B \subseteq \mathbb{N}$
2. Dismal Arithmetic (base $b \geq 2$) Let a and b be base-b numbers with digits in $\{0, 1, \ldots, b-1\}$.
   1. $a \oplus b$ is given by digitwise max (no carries).
   2. $a \otimes b$ is given by digitwise min, combined as in ordinary multiplication (no carries).
3. Numbral Arithmetic let a and b be binary sequences.
   1. $a + b$ is given by Bitwise OR.
   2. $a \times b$ is given by Shift-and-OR.
4. Polynomials in x over the Tropical semiring